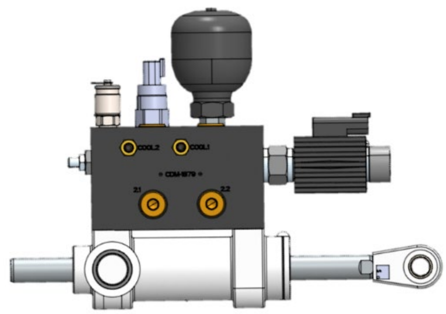
SEAKEEPER BRAKE ASSEMBLY REPLACEMENT - 211

PRODUCT SEAKEEPER 1, 4, 4.5/5, 7.5,10/10.5, AND 14

PURPOSE

This procedure guides the replacement of the brake assembly of single-cylinder Seakeeper models. These brake assemblies are not intended to be serviced in the field and therefore must be replaced as a single unit.



TOOLS/SUPPLIES REQUIRED

- Seakeeper 1 Brake Assembly (P/N 12096)
- Seakeeper 4 and 4.5/5 Brake Assembly (P/N 13104)
- Seakeeper 7.5 Brake Assembly (P/N 14530)
- Seakeeper 10/10.5 Brake Assembly (P/N 14027)
- Seakeeper 14 Brake Assembly (P/N 14133)
- Standard Metric Socket and Wrench Set
- Black Molybdenum grease
- Bushing Tool Kit: small (P/N 11367) or large (P/N 10449)
- ¼" plastic double-barbed fitting for Seakeeper 1 (McMaster-Carr 5372K513 or similar)
- ½" plastic double-barbed fitting for Seakeeper 4, 4.5/5, 7.5, 10/10.5, or 14 (McMaster-Carr 5372K518 or similar)
- Replacement hose clamps (worm-drive clamps acceptable)
  - 13/32" OD for Seakeeper 1
  - 5/8" OD for Seakeeper 4, 4.5/5, 7.5, 10/10.5, or 14
  - OPTIONAL: Pinch clamp pliers, straight & side jaw (McMaster-Carr 6541K69)
- Nickel-based Anti-seize
- Marine sealant (Sili-Thane 803 or similar)
- Thread Locker (Loctite #243 or similar)

PRECAUTIONS

1. PERSONNEL EYE SPRAY HAZARD EXISTS when working with pressurized brake assemblies. Each assembly is pressurized to 50 psi or more with AW-46 hydraulic fluid.

REFERENCES

- SWI-209 – Seakeeper Bushing Replacement
- SWI-108A – Seakeeper Angle Sensor Calibration
- SWI-118 – Seakeeper Service Tool Guide

PROCEDURE

REMOVING OLD ASSEMBLY

1. At MFD app or display, **ACTIVATE** brake override:
    a. **PRESS** wrench icon.
    b. **PRESS <u>AND</u> HOLD** wrench icon until override screen appears.
    c. **ACTIVATE** brake override (Fig. 1).

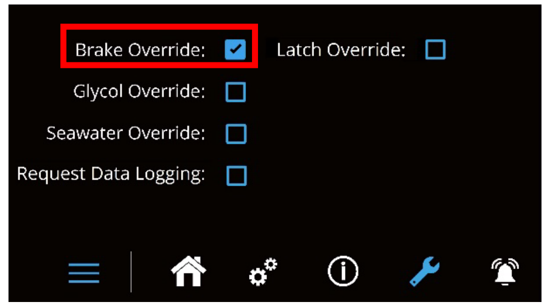


Figure 1: Override screen

2. **ACCESS** brake assembly as follows:
    a. **REMOVE** top cover.
    b. On Seakeeper 4 and above, **REMOVE** brake-side panel with 10 mm socket (Fig.3).

Figure 2: Seakeeper 1 top cover

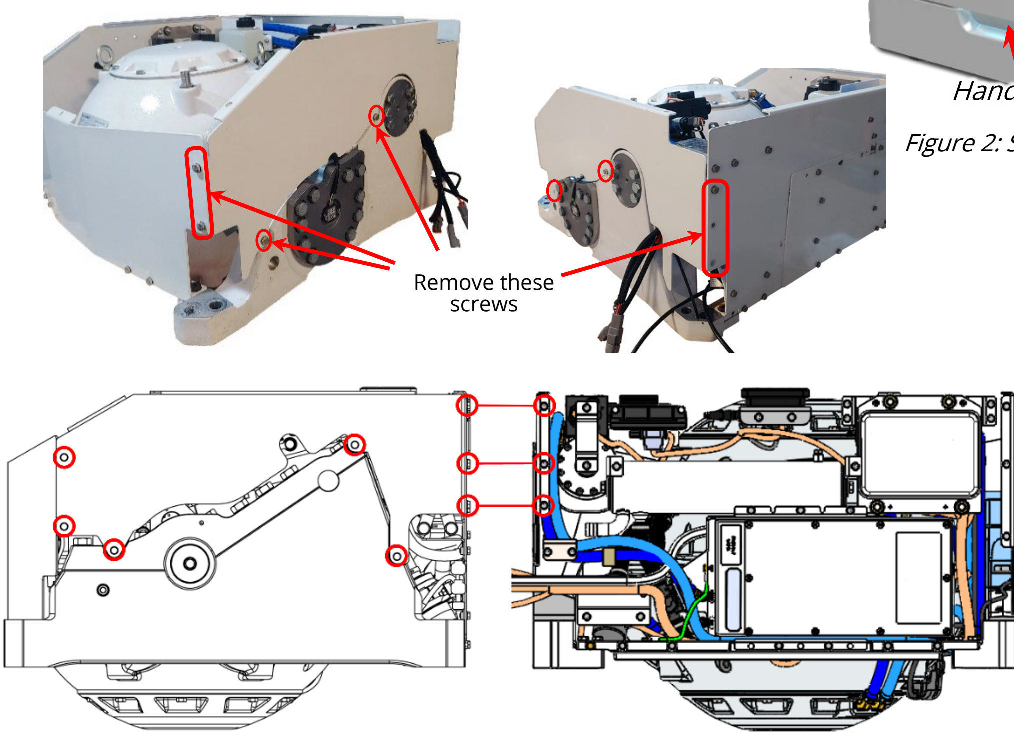


Figure 3: Seakeeper 4, 4.5/5, 10/10.5, and 14 side cover (TOP), Seakeeper 7.5 (BOTTOM)

3. **<u>IF</u>** Seakeeper 7.5,
   <u>THEN</u>:

   a. **REMOVE** seven screws of brake-side gimbal cap with a 24 and 21 mm socket.

   b. **REMOVE** brake-side gimbal cap and three gimbal cap gaskets.

4. **PRECESS** enclosure to allow access to cylinder rod-end pin (Fig. 4).

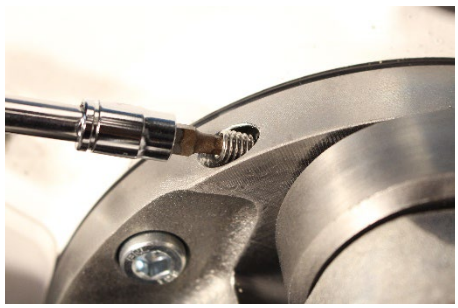
   a. **<u>IF</u>** Seakeeper 7.5,
      <u>THEN</u> ENSURE enclosure near 0° angle.

Figure 4: Seakeeper 1 positioned for access

5. **DEACTIVATE** brake override.

6. **<u>IF</u>** Seakeeper 4 or above,
   <u>THEN</u> REMOVE two brake pin set screws with 4 or 5 mm Allen hex. [Located on gimbal shaft near enclosure. See Figure 5.]

Figure 5: Seakeeper 4 through 14 set screws



WARNING: EYE INJURY MAY RESULT from snap ring potentially flying out when inserted or removed.

7. **REMOVE** internal snap retaining ring and spacer washer of rod-end pin. [Only Seakeeper 1 has a spacer washer, as shown in Figure 6]

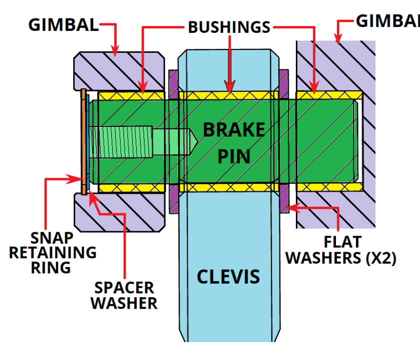


Figure 6: Seakeeper 1 rod-end details

NOTE:

Seakeeper 4, 4.5/5 have a pin blanket as shown in Figure 7c.

8. With Brake Bushing Service Tool Kit, **REMOVE** rod end pin and two washers.

Figure 7A: Brake pin retainer ring removal

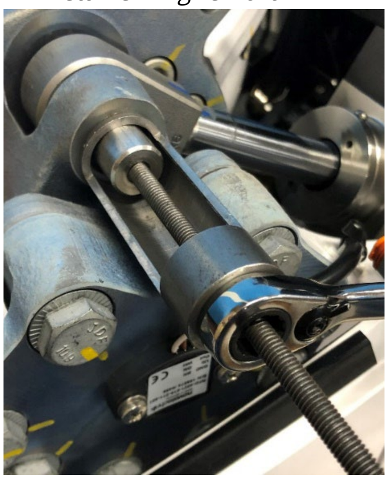


Figure 7B: Brake pin removal

Seakeeper 4/4.5/5 Details

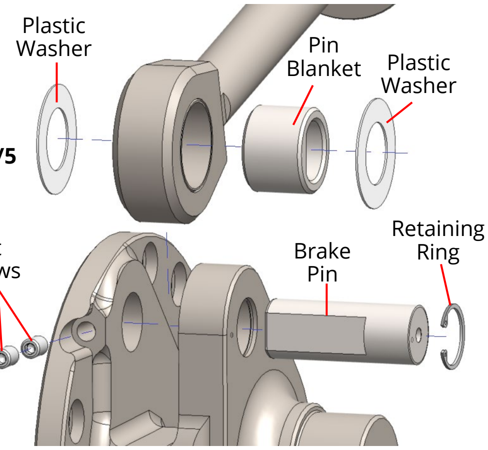


Figure 7C: Seakeeper 4 & 4.5/5 brake pin details

9. **DISCONNECT** harness connectors from brake valve and pressure switch (Fig. 8).

10. **DISCONNECT** coolant hoses at brake manifold brass barbs.

    a. **IF** hose held with pinch clamp, **THEN**:

        i. **CUT** clamp with side-cutting pliers.

        ii. **PRY** remainder of clamp from hose.

11. **CONNECT** two loose ends of coolant hoses with barbed hose fitting to minimize coolant loss per Figure 9.

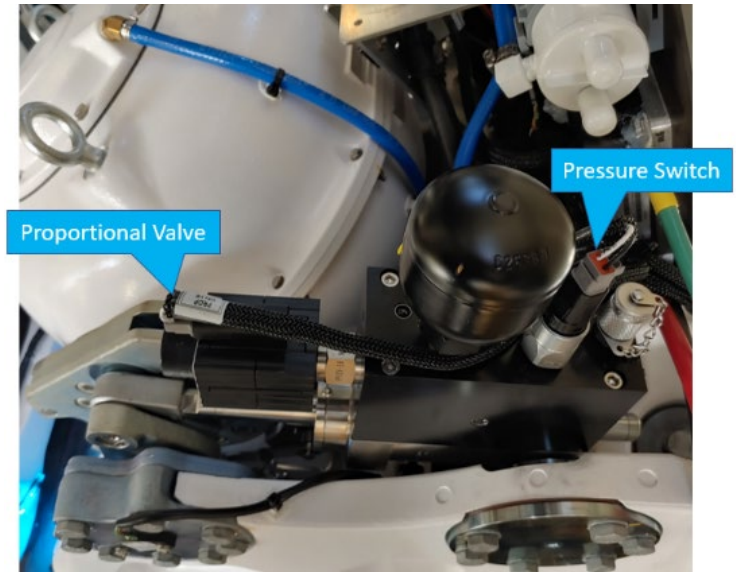


Figure 8: Disconnecting harness from manifold

    a. Seakeeper 1 uses 1/4" barbed hose fitting

    b. Seakeeper 4, 4.5/5, 7.5, 10/10.5, and 14 use 1/2" barbed hose fitting

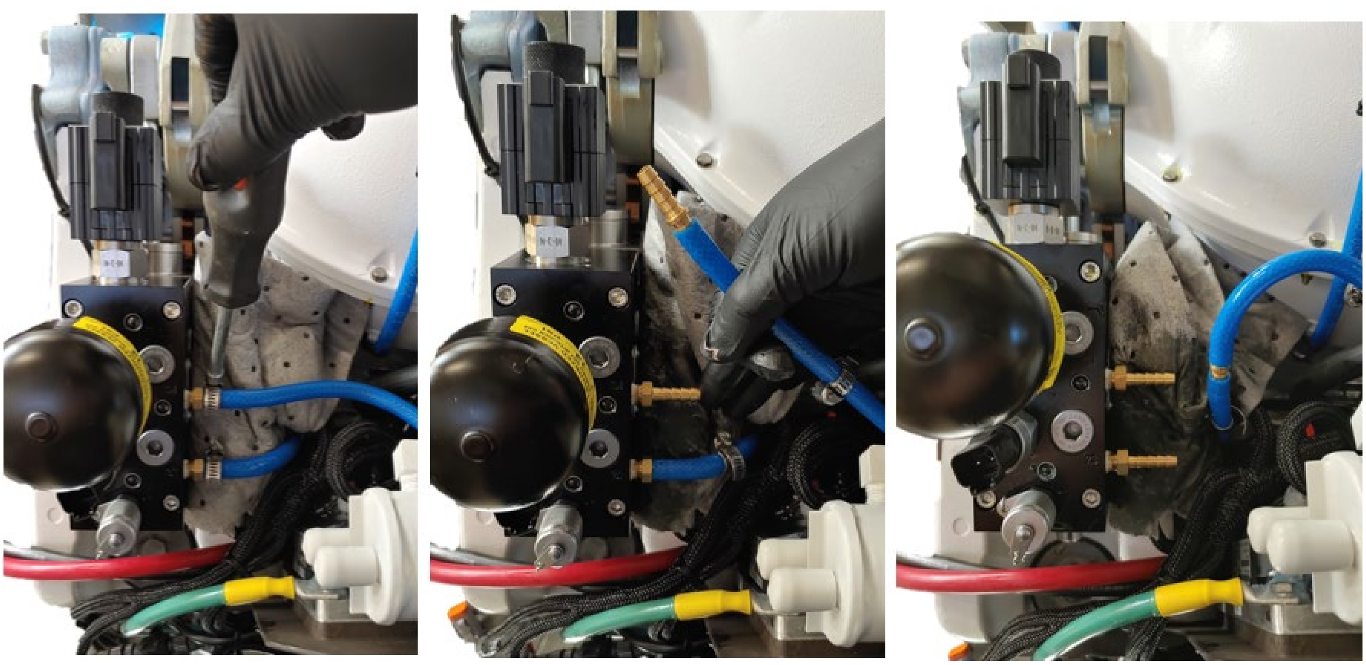


Figure 9: Removing manifold cooling lines

12. **IF** Seakeeper 1,
THEN REMOVE plastic access panel of brake assembly trunnion plate.

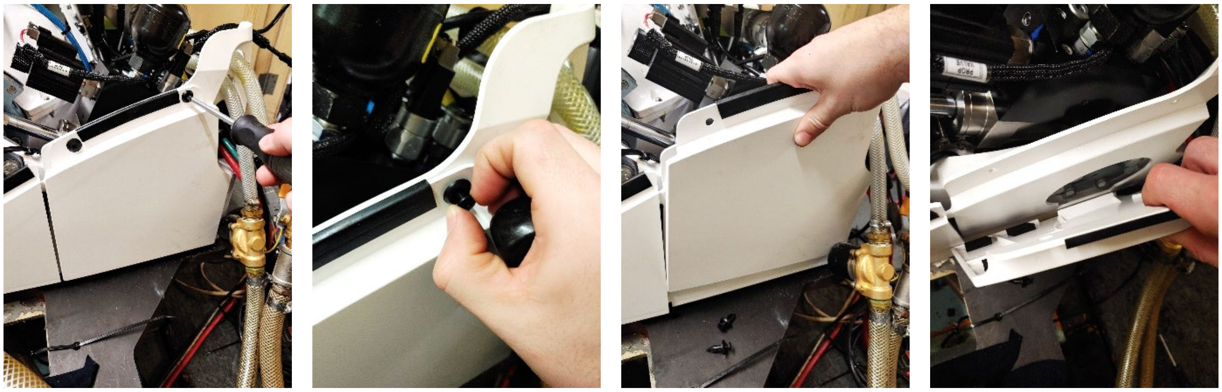


Figure 10: Removing Seakeeper 1 trunnion cover

13. **REMOVE** screws from each trunnion plate with 17 mm socket or wrench (22 mm for Seakeeper 10/10.5 and 14).

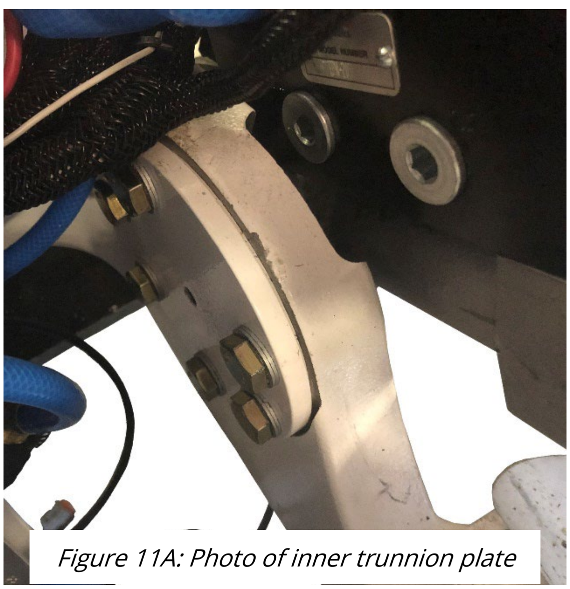


Figure 11A: Photo of inner trunnion plate

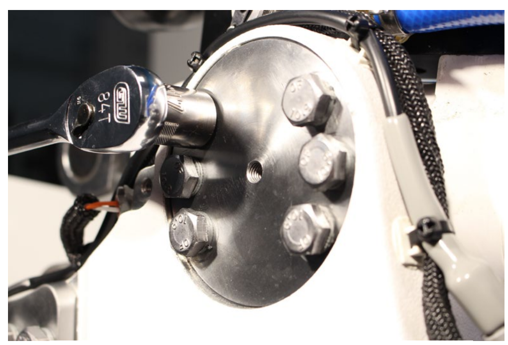


Figure 11B: Photo of outer trunnion plate of Seakeeper 4

14. **<u>IF</u>** Seakeeper 7.5,
<u>THEN</u>:

    a. **REMOVE** brake valve solenoid coil from manifold for room to remove screws.

    b. **REMOVE** three inboard trunnion cap screws with a 24 mm socket.

    c. **REMOVE** inboard trunnion cap and gasket.

15. **<u>IF</u>** Seakeeper 1, 4, 4.5/5, 10/10.5, or 14,
<u>THEN</u> REMOVE both trunnion plates.

    a. **<u>IF</u>** trunnion plate difficult to remove,
    <u>THEN</u>:

        i. **INSTALL** sourced M8-1.25 X 35 mm hex head cap jack screw, with 13 mm socket, in trunnion plate center threaded hole.

        ii. **TIGHTEN** jack screw until trunnion plate free of brake assembly trunnion and foundation (Fig. 12).

16. **REMOVE** brake assembly from Seakeeper as follows:

    a. **SUPPORT** weight of assembly during next step to avoid dropping.

    b. **REMOVE** brake assembly from Seakeeper.

    c. **SET** brake assembly aside.

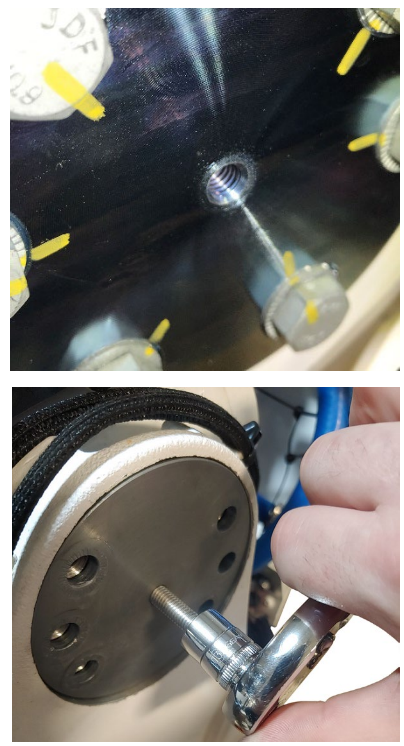


Figure 12: TOP – jack screw threaded hole in trunnion plate. BOTTOM – Jack screw used to remove trunnion plate

INSTALLING REPLACEMENT ASSEMBLY

17. **IF** Seakeeper 1, 4, 4.5/5, 10/10.5, or 14,
 THEN:

 a. **APPLY** light film of black molybdenum or vacuum grease to trunnion plate O-rings.

 b. **INSTALL** O-rings onto trunnion plates.

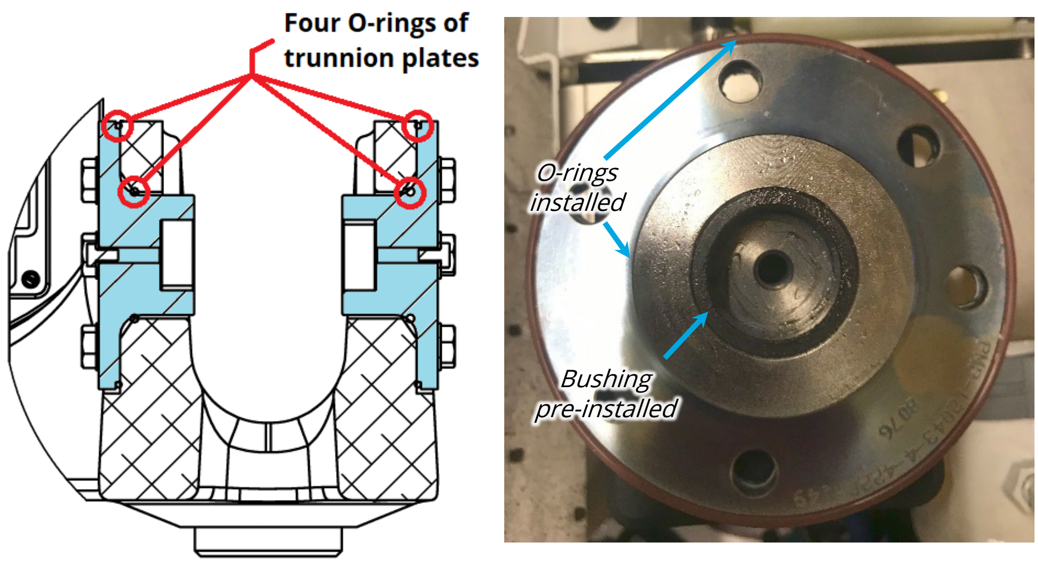


Figure 13: Seakeeper 1 Details Shown (Seakeeper 4, 4.5/5, 10/10.5, and 14 similar)

18. **PREPARE** trunnion or gimbal cap screws for installation:

 a. **IF** reusing fasteners,
 THEN:

 i. **CLEAN** screw heads and wedge lock washers of old sealant.

 ii. **LEAN** threads of old anti-seize compound.

 b. **APPLY** anti-seize to threads.

 c. **IF** Seakeeper 1, 4, 4.5/5, 10/10.5, or 14,
 THEN **APPLY** marine sealant to under screw heads and wedge lock washers.

Figure 14: Properly coated trunnion screw

19. **IF** Seakeeper 1, 4, 4.5/5, 10/10.5, or 14,
THEN INSTALL trunnion plates as follows:

    a. **INSTALL** plastic (PTFE) flat washers on brake assembly trunnions.

    b. **PLACE** replacement brake assembly into Seakeeper.

    c. **ALIGN** inner trunnion axle of cylinder with inner trunnion plate.

    d. **INSERT** inner trunnion plate over inner trunnion axle of cylinder

    e. **ALIGN** inner trunnion plate screw holes with foundation holes.

    f. **HAND-TIGHTEN** inner trunnion plate fasteners to hold brake assembly.

    g. **INSTALL** outer trunnion plate with fasteners.

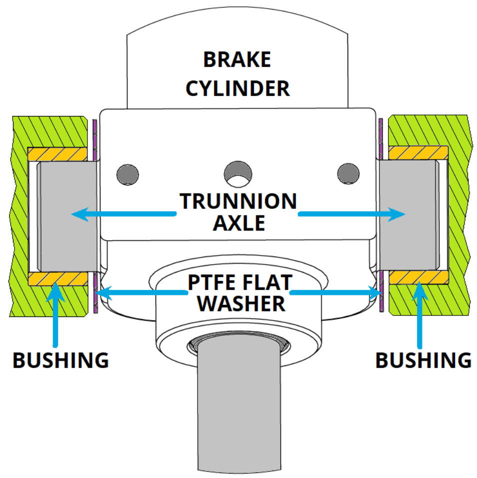


Figure 15: Trunnion axle details

20. **IF** Seakeeper 7.5,
THEN:

    a. **POSITION** inboard gimbal cap gasket (P/N 14195) on Seakeeper per Figure 16.

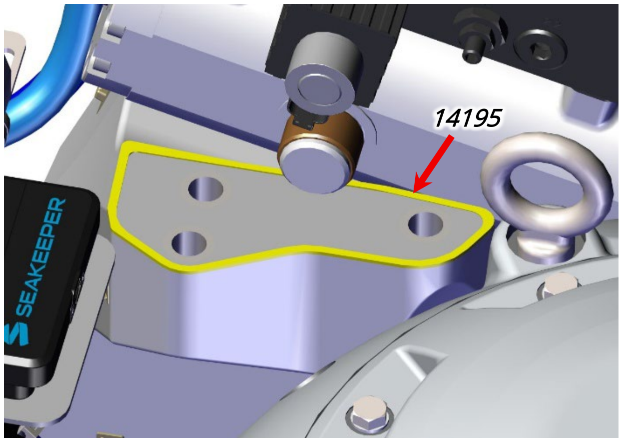


Figure 16: Inboard gimbal cap gasket detail.

    b. **PLACE** plastic washers over brake assembly trunnions.

    c. **PLACE** inboard gimbal cap over inboard trunnion of brake assembly (Fig 17).

    d. **PLACE** trunnion sleeve (P/N 14113) with O-rings over outboard trunnion of brake assembly (Fig. 17).

    e. **POSITION** replacement brake assembly in Seakeeper.

    f. **POSITION** brake-side gimbal cap

    g. **HAND-TIGHTEN** three inboard trunnion cap M16-2 X 80 mm HHC screws and washers (P/N 61321 & 61284)

21. **TORQUE** both trunnion plate screws:

    a. Seakeeper 1, 4, and 4.5/5: **40 ft-lbs** (**54 Nm**)

    b. Seakeeper 7.5 inboard trunnion cap: **150 ft-lbs (203 Nm)**

    c. Seakeeper 10/10.5: **70 ft-lbs** (**95 Nm**)

    d. Seakeeper 14: **79 ft-lbs** (**107 Nm**)

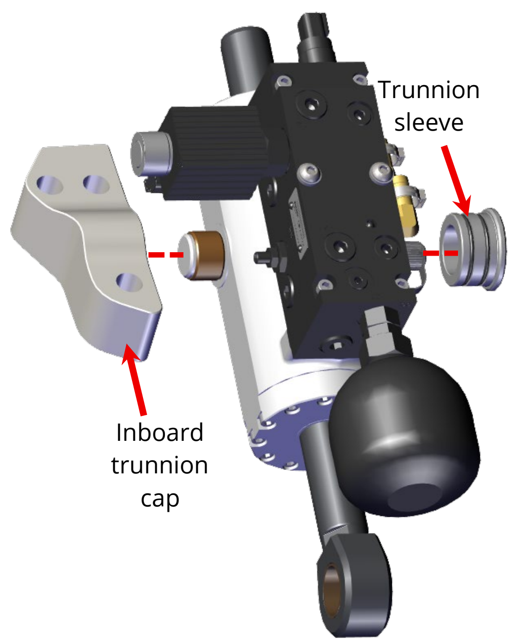


Figure 17: Brake assembly detail

22. **IF** Seakeeper 4 or 4.5/5,
THEN APPLY thin film of back molybdenum grease to inside and outside surfaces of pin blanket (Fig. 18).

23. **APPLY** film of black molybdenum grease to rod-end brake pin.

24. **IF** rod-end clevis does not align with brake pin opening, **THEN**:

    a. **CONNECT** brake valve harness connector to assembly's brake valve.

    b. **ACTIVATE** brake override.

    c. **POSITION** cylinder rod clevis as needed.

    d. **DEACTIVATE** brake override.

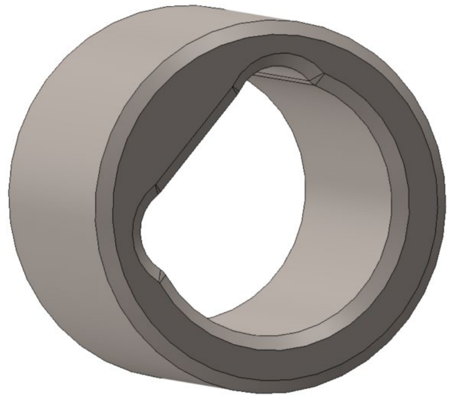


Figure 18: Seakeeper 4/4.5/5 Pin Blanket

25. With threaded hole facing out, **INSTALL** rod-end brake pin through rod end clevis and gimbal with two flat washers as shown below.
[**NOTE**: Brake pin alignment dimples must be aligned for set screws to engage pin correctly.]

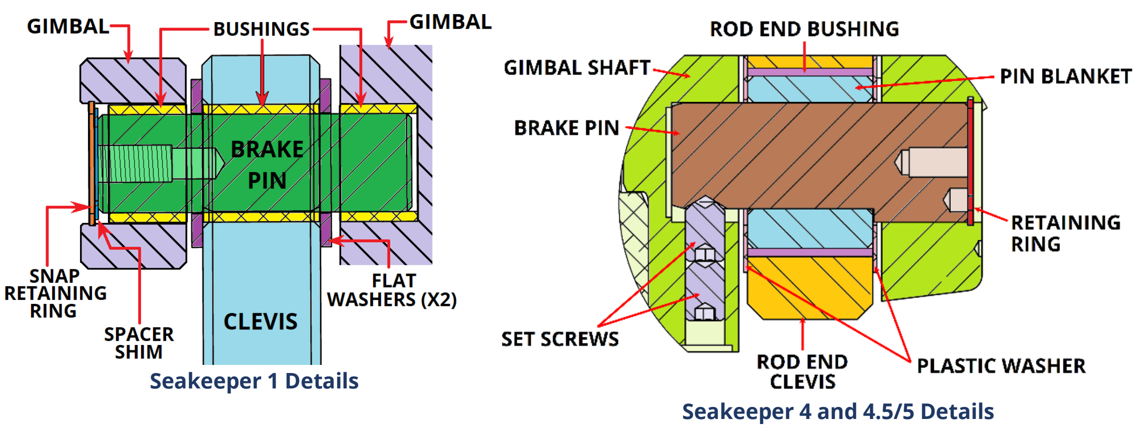


Figure 19A: Brake pin details

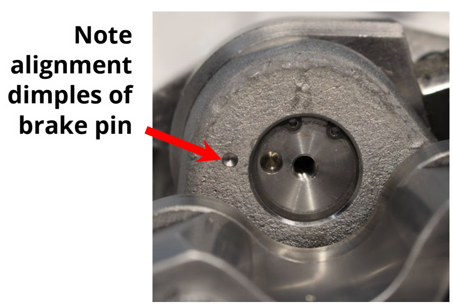


Figure 19B: Brake pin and gimbal alignment dimples

26. <u>**IF**</u> Seakeeper 4 or above,
<u>**THEN**</u> **INSTALL** set screws as follows:

    a. **APPLY** blue thread locker to threads of two set screws.

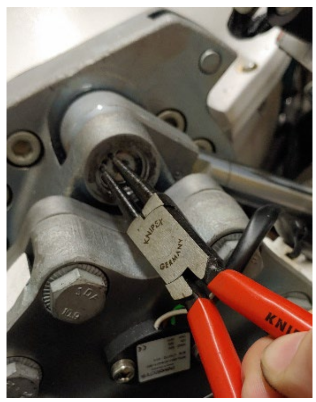
    b. **INSTALL** set screws to secure brake pin with 4 or 5 mm Allen hex.

    c. **TORQUE** each set screw to **26 ft-lbs** (**35 Nm**).

WARNING:
EYE INJURY MAY RESULT from snap ring potentially flying out when inserted or removed.

27. **INSTALL** spacer shim (Seakeeper 1 only) and internal snap retaining ring (Fig. 20).

Figure 20: Brake pin retainer ring installation



28. <u>**IF**</u> Seakeeper 1,
<u>**THEN**</u> **INSTALL** outer trunnion access cover with hardware.

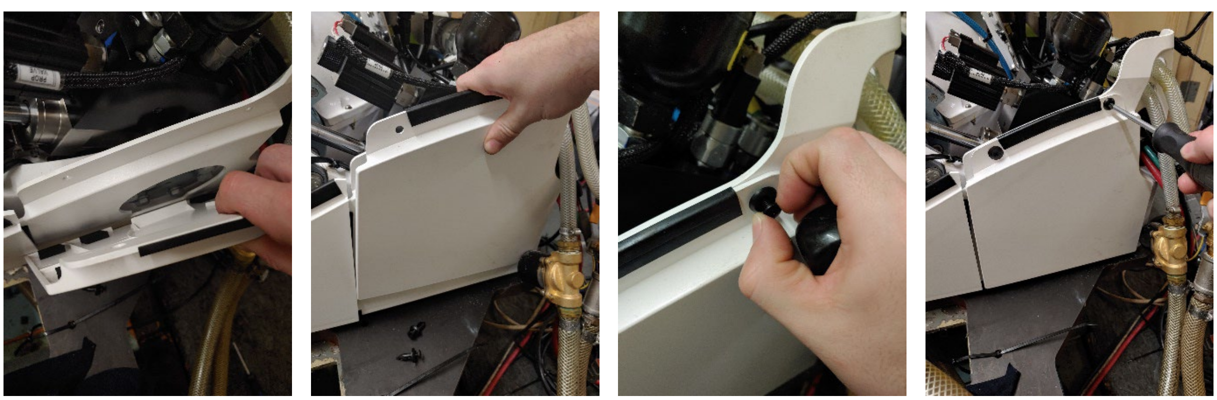


Figure 21: Seakeeper 1 trunnion plate installation

29. **IF** Seakeeper 7.5,
    THEN:

    a. **INSTALL** three gaskets of brake-side gimbal cap per Figure 22.

    b. **POSITION** brake-side gimbal cap onto Seakeeper frame.

    c. **HAND-TIGHTEN** gimbal cap screws per Figure 23.

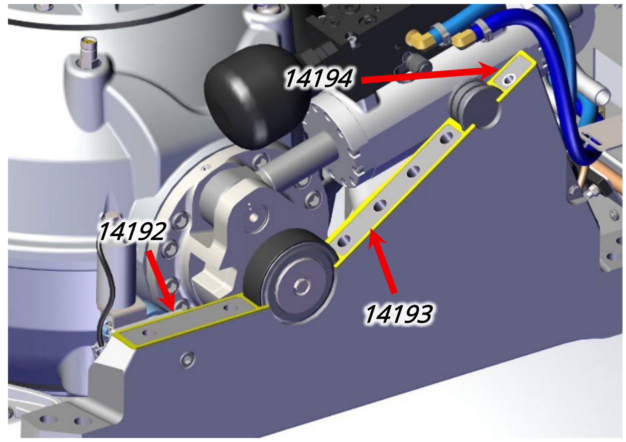


Figure 22: Brake-side gimbal cap gaskets

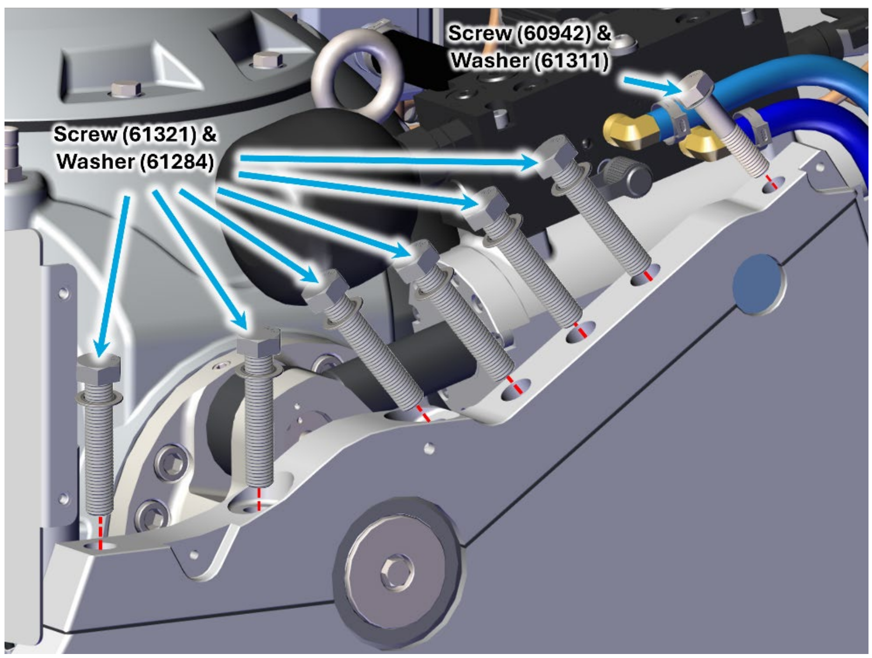


Figure 23: Seakeeper 7.5 gimbal cap details with part nos.

    d. **TORQUE** six M16 HHC screws to **150 ft-lbs (203 Nm)** with 24 mm socket.

    e. **TORQUE** rear M12 HHC screw to **75 ft-lbs (102 Nm)** with 21 mm socket.

30. **RECONNECT** wire harness connectors to brake valve and manifold pressure switch.

31. **REMOVE** coolant hoses from barbed fitting.

NOTE:
Coolant hose pinch clamps cannot be reused.

32. **REINSTALL** coolant hoses on brake manifold fittings:

    a. **REPLACE** used pinch clamps with worm-drive clamps.

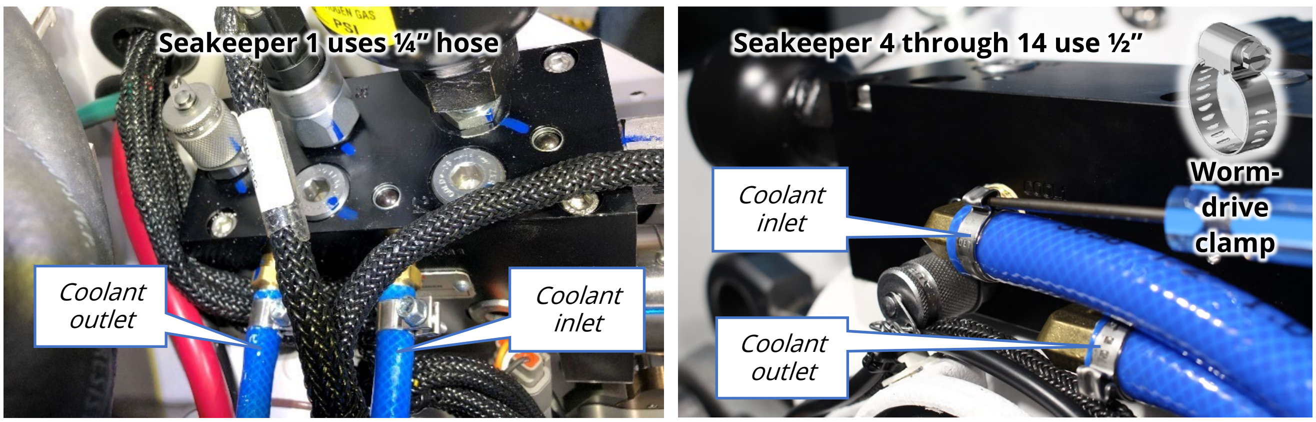


Figure 24: Manifold cooling line installation

33. **WIPE** any spilled glycol coolant.

34. **ENSURE** coolant reservoir level satisfactory.

35. **INSTALL** cable ties where cut for brake assembly replacement.

36. **ENSURE** angle sensor is calibrated.

    a. **IF** angle sensor requires calibration, **THEN PERFROM** SWI-108A to calibrate angle sensor.

37. **INSTALL** covers removed.

38. **IF** Seakeeper 4 or 4.5/5, **THEN**:

    a. **INSTALL** Seakeeper Service Tool app on Seakeeper per **SWI-118 – Seakeeper Service Tool Guide**.

    b. **ENTER** new PVMO value of **1615** (Fig. 25).

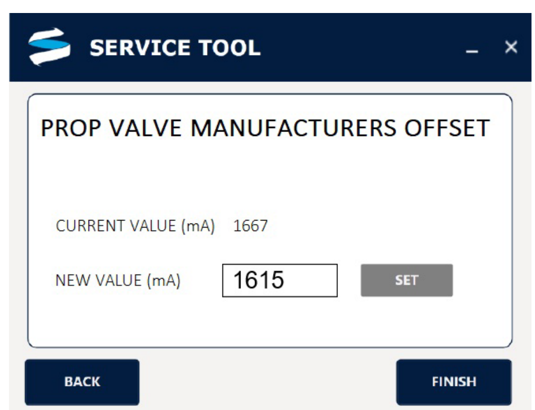


Figure 25: PVMO screen in Service Tool

39. **ADDRESS** any alarms that may be on MFD app / display.

40. **PERFORM** sea trial to ensure operability.

********** END **********

Revision	Description	Approval	Date
6	Seakeeper 14 introduction. Minor edits throughout.	A Patricio	23JUL2025
7	Added step to reposition cylinder rod if not aligned properly. Seakeeper 5, 7.5, and 10.5 models introduced.	A Patricio	23JUL2026